

HR Analytics | People Analytics | Power BI | SQL | Python

1,480

Employees
Analyzed

238

Employees
Left

16.1%

Attrition
Rate

39.3%

Sales Rep
Attrition

30.6%

Overtime
Attrition

\$27.5M

Est. Annual
Replacement

Key Attrition Findings

Driver	Finding	Attrition Rate	Business Implication
Job Role	Sales Representatives at highest risk	39.3%	Specialized retention program needed immediately
Overtime	Employees working overtime churn 3x more	30.6% vs 10.4%	Overtime policy review — highest leverage action
Age Group	Youngest employees (18-25) most likely to leave	35.8%	Structured early career development program required
Salary	Sub-\$5K employees churn at 21.6% — highest band	21.6% vs 3.8%	Compensation review for entry-level roles
Department	Sales dept churns faster than R&D	20.7% vs 13.8%	Sales-specific manager training and incentives
Marital Status	Single employees leave at 2x the rate of divorced	25.4% vs 10.4%	Early career flexibility and community building
Travel	Frequent travelers churn 3x more than non-travelers	24.7% vs 7.9%	Travel policy cap and remote work alternatives

Business Recommendations

Priority	Recommendation	Target Segment	Impact
1 (Critical)	Overtime policy: cap mandatory OT, add compensation	All employees on OT	Reduce 30.6% OT attrition by 40-50%
2 (Critical)	Sales Rep retention program: structured career path + sales tips	Sales Reps (20.7% rate)	Reduce Sales attrition below 15%
3 (High)	Salary bands review for sub-\$5K employees	753 employees at risk	Reduce 21.6% low-salary attrition
4 (High)	18-25 mentorship + early career development	123 young employees	Reduce 35.8% attrition in youngest cohort
5 (Medium)	Frequent travel cap + remote work flexibility	Frequent travelers	Reduce 24.7% travel-related attrition
6 (Medium)	Exit interview program to validate data findings	All departing staff	Improve diagnosis accuracy

Financial context: At an industry-standard replacement cost of 2x monthly salary, the 238 annual departures cost an estimated **\$27.5M per year**. Even reducing attrition by 30% saves approximately \$8.2M annually.

Technology Stack & Skills Demonstrated

Category	Tools & Skills
BI Dashboard	Power BI — KPI Cards, Dept/Role/Age/Salary breakdowns, Interactive slicers
Python	Pandas, NumPy, Matplotlib, Seaborn — EDA, feature correlation, 9 visual outputs
SQL	CTEs, Window Functions (RANK, DENSE_RANK, LAG, NTILE), CASE WHEN, JOINS, aggregations
Data Skills	HR Analytics, People Analytics, Employee Attrition, Workforce Segmentation, KPI Reporting
Business Skills	Stakeholder Reporting, Business Intelligence, Executive Summaries, ROI Analysis
Tools	Excel, Jupyter Notebook, Git, GitHub, Markdown, PDF Reporting

Built for: HR Analyst | Data Analyst | BI Analyst | Reporting Analyst | People Analytics | Remote Analytics Roles